

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		mass of copper produced = 1.39g (1) second mark awarded for working showing yield to be 109% $\text{yield} = \frac{1.39}{1.27} \times 100 = 109\%$ (1) no ecf possible		2		2	2	2
		(ii)		the copper(II) oxide / charcoal was impure / some charcoal reacts with the air / not all the copper(II) oxide reacted / charcoal in excess accept 'not enough charcoal to reduce all of the copper(II) oxide' reference to heat or time is neutral			1	1		1
	(b)	(i)		accept reduction by carbon monoxide or carbon $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ $2\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Fe} + 3\text{CO}_2$ reactants and products (1) balancing (1) balancing mark can only be awarded if both the reactants and products are correct	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		7.7×10^4 (3) 77000 (2) if answer incorrect credit each of following $\frac{22}{100} \times (5 \times 10^5)$ or 1.1×10^5 (1) $\frac{112}{160} \times (1.1 \times 10^5)$ or $0.7 \times (1.1 \times 10^5)$ (1) ecf possible		3			3	
				Question 5 total	2	5	1	8	5	3